Unifying CEU's Course Information System

A Student-Centered Database Redesign — Prototype

David Diestre Rubio • Teodore Giorgobiani • Thanika Haltrich • Talha Sadiq

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Presentation Overview

~5 min

Prototype

- Research problem
- Hypothesis
- ER diagram
- Live queries
- GitHub

~5 min

Progress Report

- Implementation vs. proposal
- Timeline feasibility
- Best practices learned

~5 min

Future Plan

- Next 4 weeks
- New additions
- Updated goals

Section I: Prototype

Why, What, and How

The Research Problem

Problem Statement

- Info across 3 platforms
- No unified access point
- Manual schedule building
- Old codes break search
- Cross-dept courses invisible

Student Impact

- Platforms feel disconnected
- Course descriptions often missing
- Students miss eligible courses
- Everyone rebuild schedules manually in Timeedits
- Old codes make courses unfindable

What We Want to Achieve

- *One unified database*
- *Replaces fragmented infrastructure*

Unified View

- All departments, one place
- Cross-registrations included

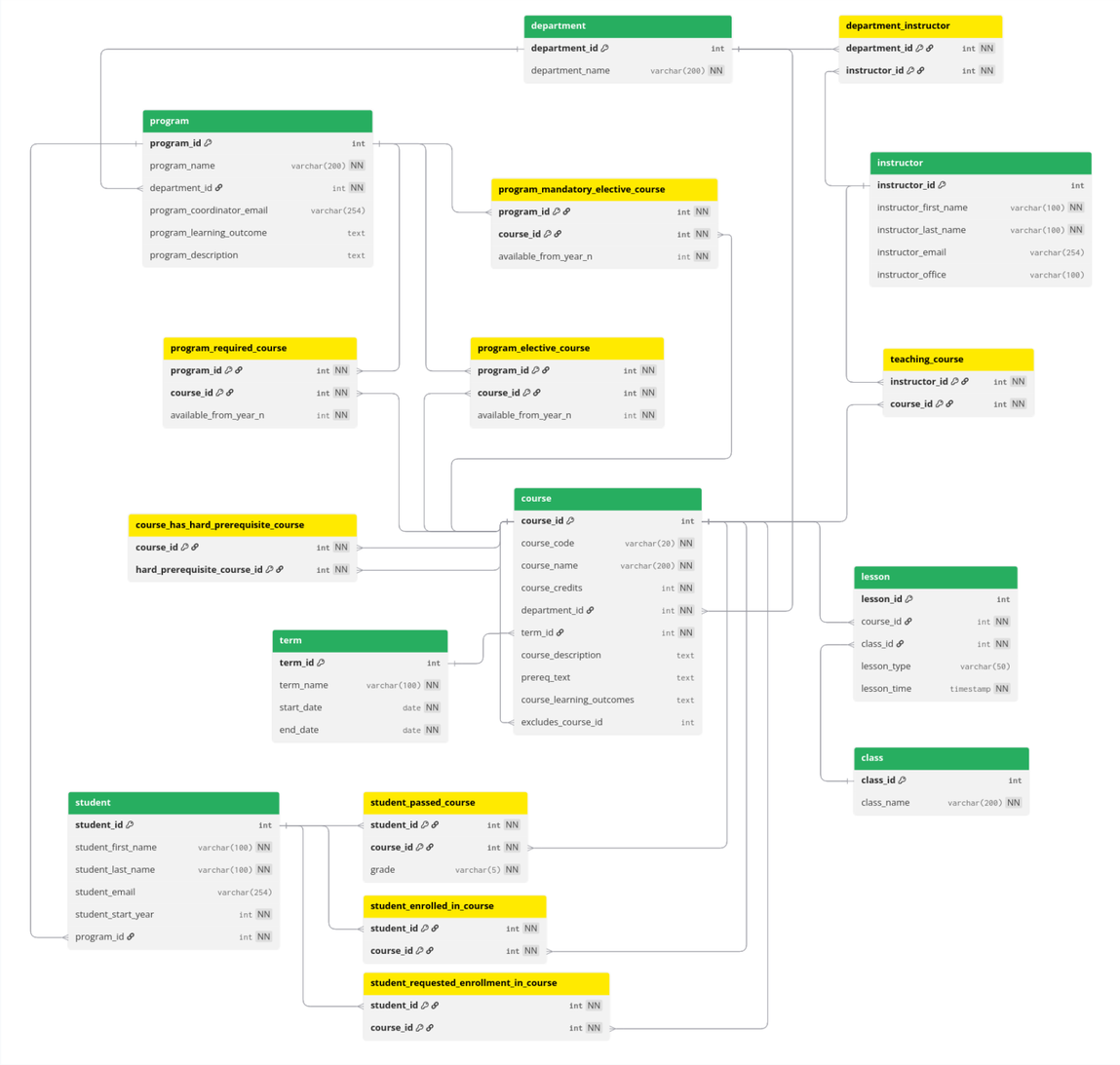
Smart Filtering

- Prerequisite-aware course view
- Filters passed/enrolled courses

Better Planning

- Track credit progress
- Plan terms easily

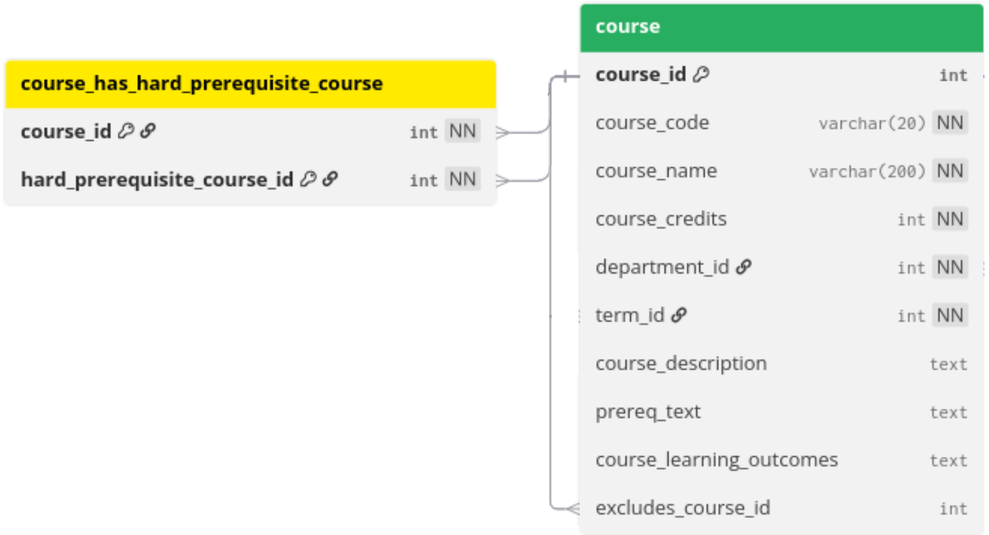
Database Design — ER Diagram

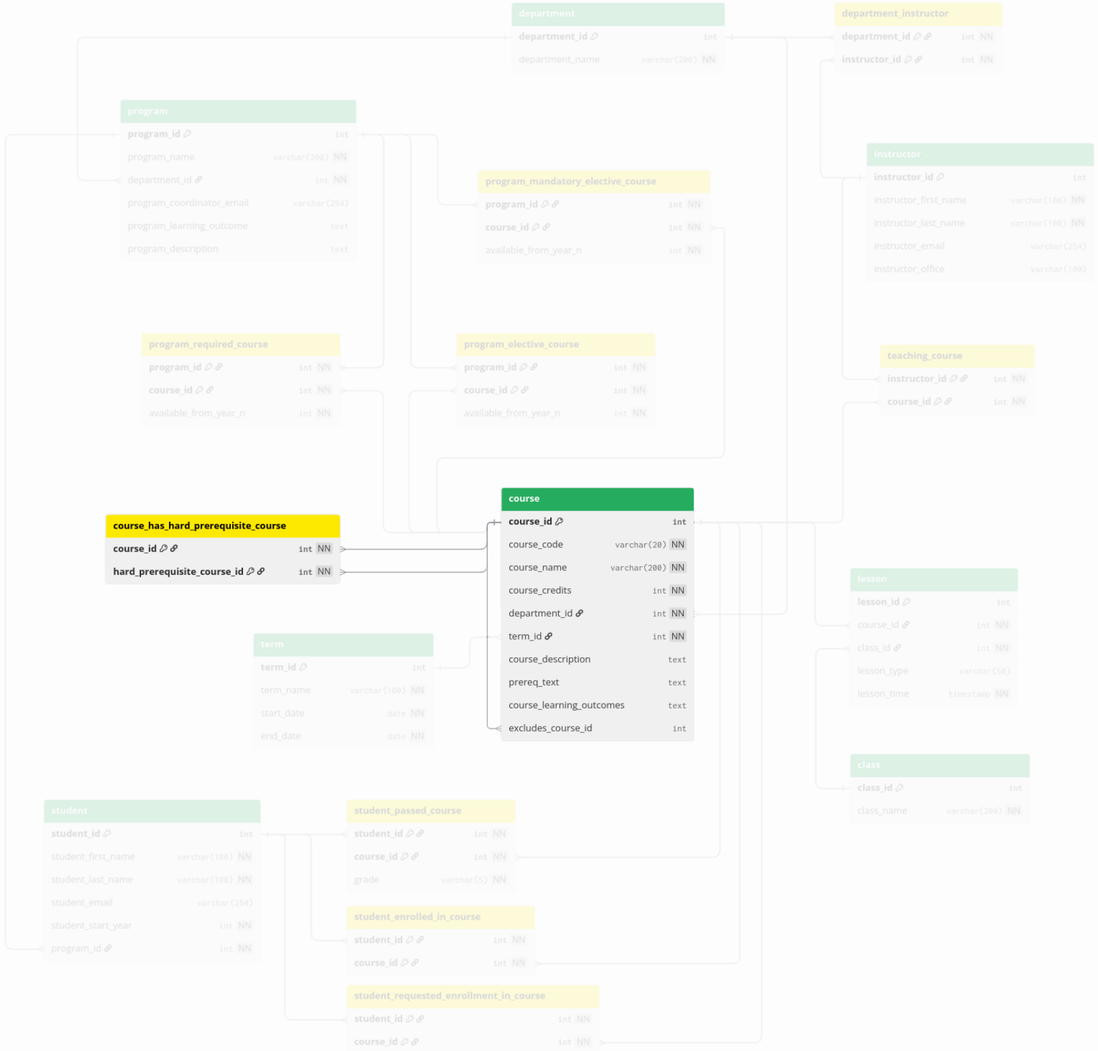


Database Design — ER Diagram

Core Entities

course





Database Design — ER Diagram

Core Entities

course

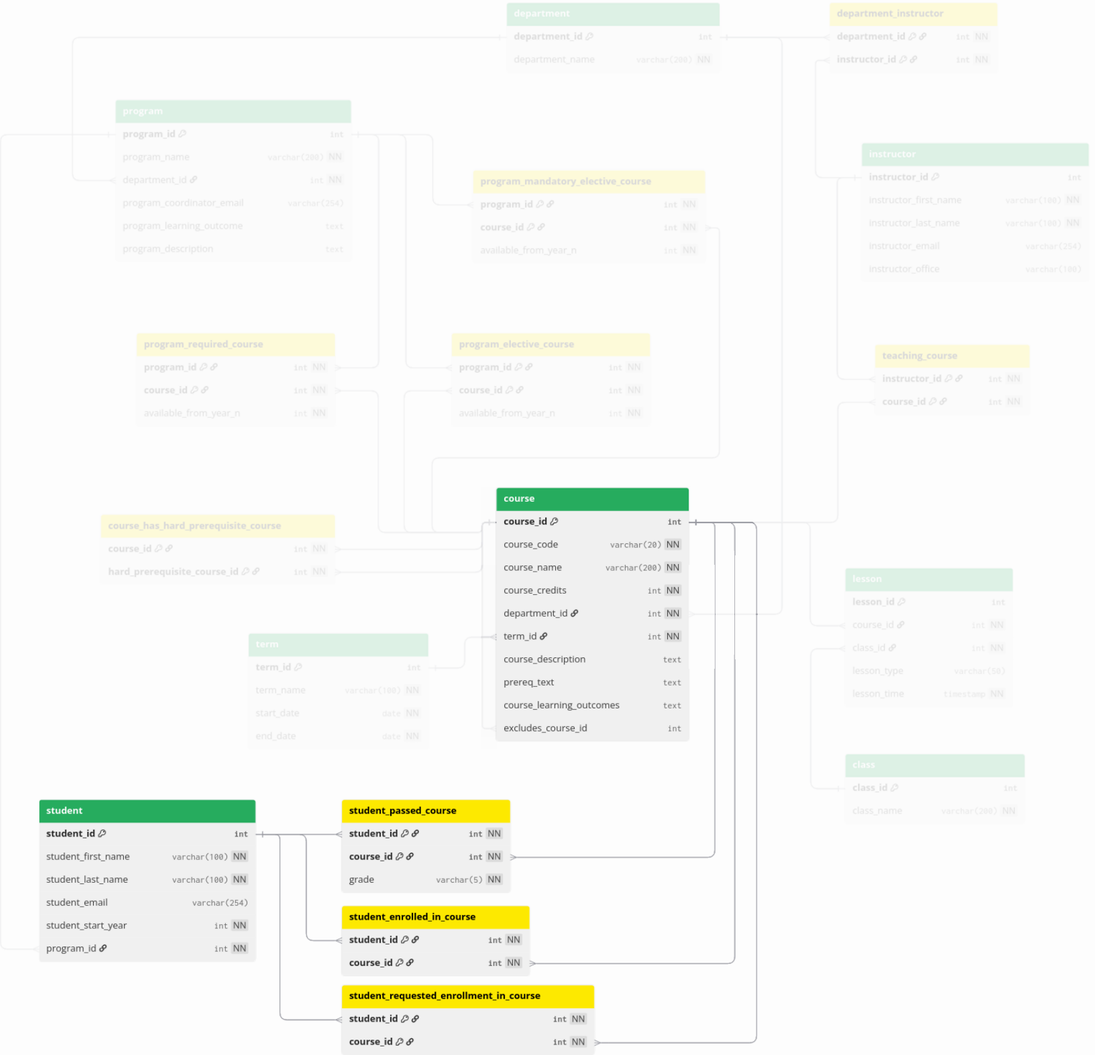
student

Junction Tables (student - course)

student_passed_course

student_enrolled_in_course

student_requested_enrollment_in_course



Database Design — ER Diagram

Core Entities

course

student

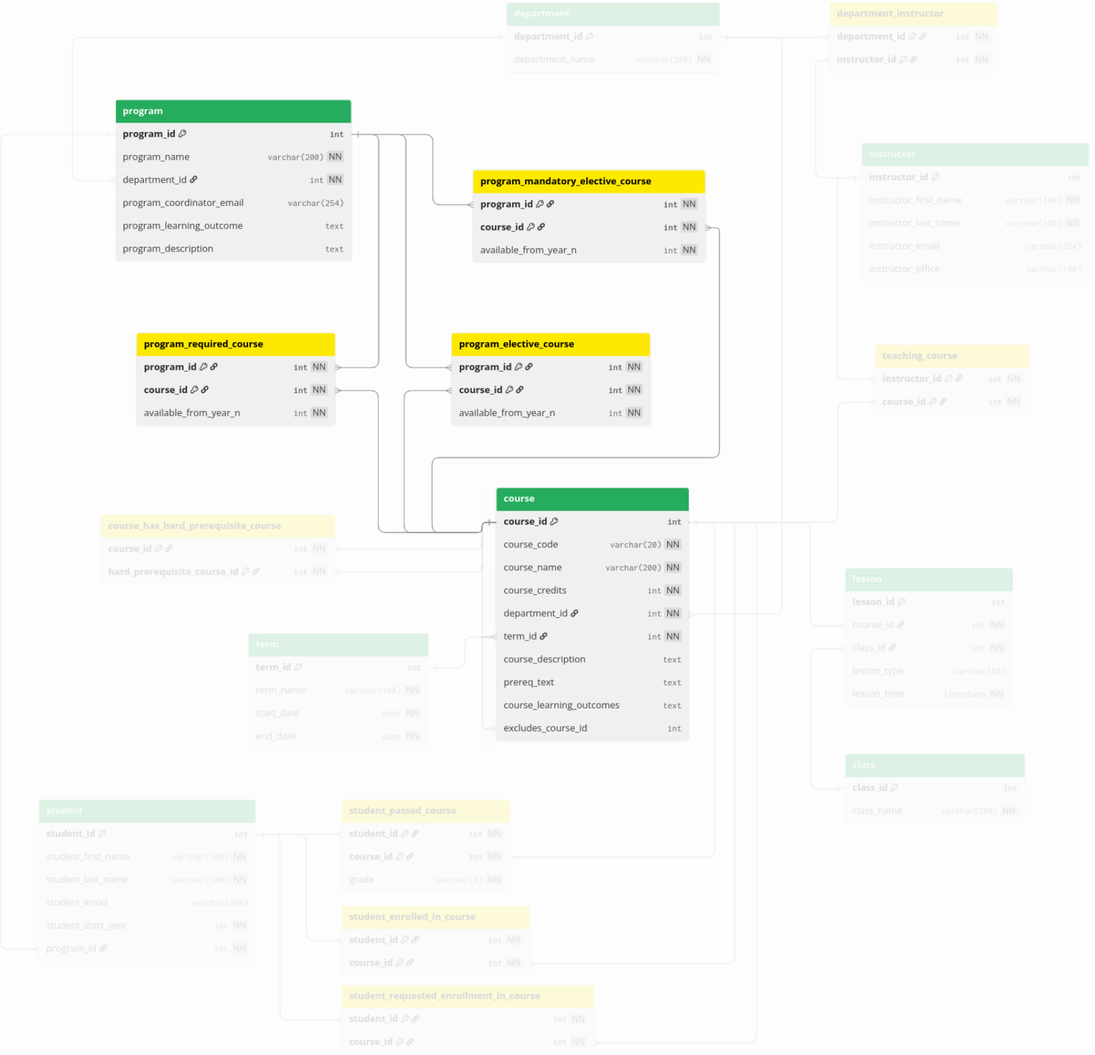
program

Junction Tables (program - course)

program_required_course

program_mandatory_elective_course

program_elective_course



Database Design — ER Diagram

Other Entities

department

instructor

lesson

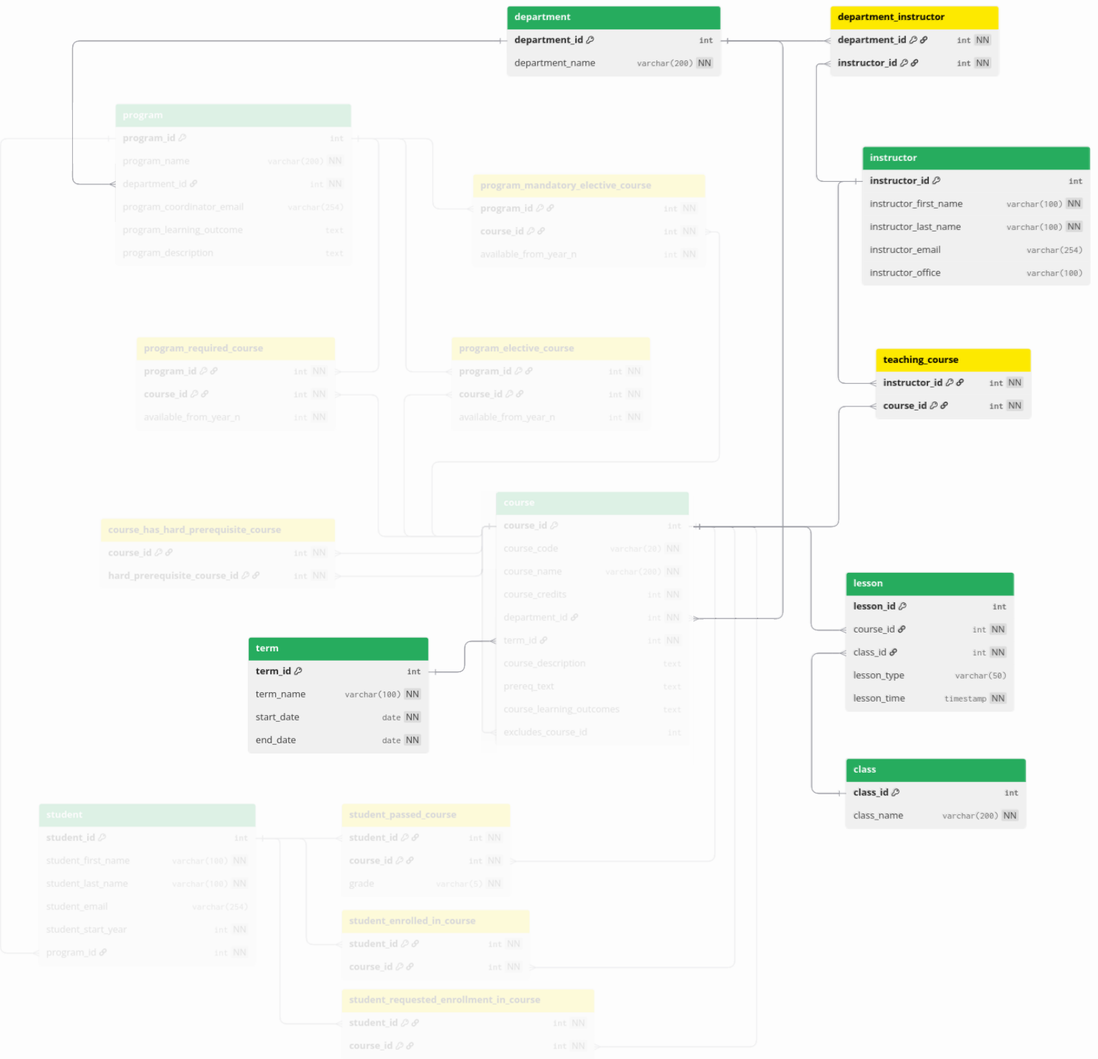
term

class

Other Junction Tables

department_instructor

teaching_course



Sample Queries & Demo

Credit Calculation

```sql
-- example: mandatory credit progress per student
SELECT s.student_last_name,
       p.program_name,
       SUM(c.course_credits)        AS total,
       COALESCE(SUM(c.course_credits) FILTER (WHERE spc.student_id
                IS NOT NULL), 0)                                  AS passed,
       SUM(c.course_credits) - COALESCE(SUM(c.course_credits)
           FILTER (WHERE spc.student_id IS NOT NULL), 0)          AS remaining
  FROM student s
  JOIN program p                   ON s.program_id  = p.program_id
  JOIN program_required_course prc ON p.program_id  = prc.program_id
  JOIN course c                    ON prc.course_id = c.course_id
  LEFT JOIN student_passed_course spc
         ON spc.student_id = s.student_id
        AND spc.course_id  = c.course_id
 GROUP BY s.student_id, p.program_name
 ORDER BY s.student_last_name;
```

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  JOIN course c                    ON prc.course_id = c.course_id
  LEFT JOIN student_passed_course spc
         ON spc.student_id = s.student_id
        AND spc.course_id  = c.course_id
 GROUP BY s.student_id, p.program_name
 ORDER BY s.student_last_name;
```

student_name	program_name	total_mandatory_credits	passed_mandatory_credits	remaining_mandatory
Ava Fischer	BSc Business Analytics	5	0	5
Emma Evans	BSc Business Analytics	5	5	0
Ethan Ionescu	BSc Computer Science	23	6	17
Liam Dobrev	MSc Data Science	12	6	6
Lucas Garcia	MSc Applied AI	12	0	12
Mia Anders	BSc Computer Science	23	6	17
Noah Balogh	BSc Computer Science	23	11	12
Olivia Chen	MSc Data Science	12	6	6
Sofia Horvath	MSc Applied AI	12	0	12
Zoe Jensen	MSc Data Science	12	0	12

Sample Queries & Demo

Course Planning

```sql
-- example: available courses for a given student
SELECT c.course_code,
       c.course_name,
       'Required'   AS course_type
  FROM student s
  JOIN program_required_course prc USING (program_id)
  JOIN course c                   USING (course_id)
 WHERE s.student_id = 2
   AND c.course_id NOT IN (SELECT course_id
                           FROM student_passed_course
                          WHERE student_id = s.student_id)
 ORDER BY course_type, course_code;
-- (UNION ALL with elective & mandatory-elective omitted for brevity)
```

Sample Queries & Demo

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  FROM student s
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  JOIN course c                   USING (course_id)
 WHERE s.student_id = 2
   AND c.course_id NOT IN (SELECT course_id
                             FROM student_passed_course
                            WHERE student_id = s.student_id)
 ORDER BY course_type, course_code;
-- (UNION ALL with elective & mandatory-elective omitted for brevity)
```

course_code	course_name	course_credits	term_name	course_type	available_from_year_n
BA201	Business Intelligence	5	Spring 2026	Elective	3
DS101	Statistics for Data Science	6	Fall 2025	Elective	2
BA201	Business Intelligence	5	Spring 2026	Mandatory Elective	2
CS301	Database Systems	6	Summer 2026	Required	3

Sample Queries & Demo

Enrollment Logic

```sql
-- example: auto-approve or flag pending enrollment requests
SELECT s.student_last_name,
       c.course_code,
       COALESCE(STRING_AGG(prereq.course_name, ', '), 'None') AS hard_prerequisites,
       CASE
           WHEN COUNT(chpc.hard_prerequisite_course_id) = 0           THEN 'AUTO-APPROVE'
           WHEN COUNT(spc.course_id) = COUNT(chpc.hard_prerequisite_course_id) THEN 'AUTO-APPROVE'
           ELSE 'FLAG: prerequisites not met'
       END AS decision
  FROM student_requested_enrollment_in_course req
  JOIN student s ON req.student_id = s.student_id
  JOIN course  c ON req.course_id  = c.course_id
  LEFT JOIN course_has_hard_prerequisite_course chpc ON chpc.course_id = c.course_id
  LEFT JOIN course prereq ON prereq.course_id = chpc.hard_prerequisite_course_id
  LEFT JOIN student_passed_course spc
         ON spc.student_id = req.student_id
        AND spc.course_id  = chpc.hard_prerequisite_course_id
 GROUP BY req.student_id, req.course_id,
          s.student_last_name, c.course_code
 ORDER BY decision, s.student_last_name;
```

Sample Queries & Demo

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         ON spc.student_id = req.student_id
        AND spc.course_id  = chpc.hard_prerequisite_course_id
 GROUP BY req.student_id, req.course_id,
          s.student_last_name, c.course_code
 ORDER BY decision, s.student_last_name;
```

student_name	course_code	course_name	hard_prerequisites	decision
Mia Anders	CS201	Data Structures	Programming Fundamentals	AUTO-APPROVE
Olivia Chen	DS201	Machine Learning	Statistics for Data Science	AUTO-APPROVE
Liam Dobrev	DS201	Machine Learning	Statistics for Data Science	AUTO-APPROVE
Emma Evans	BA201	Business Intelligence	None	AUTO-APPROVE
Ava Fischer	BA201	Business Intelligence	None	AUTO-APPROVE
Noah Balogh	CS301	Database Systems	Data Structures	FLAG: prerequisites not met
Lucas Garcia	AI301	Applied AI Systems	Data Structures, Machine Learning	FLAG: prerequisites not met
Sofia Horvath	AI301	Applied AI Systems	Data Structures, Machine Learning	FLAG: prerequisites not met
Ethan Ionescu	CS201	Data Structures	Programming Fundamentals	FLAG: prerequisites not met
Zoe Jensen	DS201	Machine Learning	Statistics for Data Science	FLAG: prerequisites not met

GitHub Repository

```
CEU-DMDB-GroupH/
├─ README.md
├─ .gitignore
├─ .env
├─ Makefile
├─ sql/
│  ├─ 00_reset.sql
│  ├─ 01_schema.sql
│  ├─ 02_constraints.sql
│  ├─ 03_views.sql
│  ├─ 04_load_from_dir.sql
│  └─ queries/
│     ├─ example_queries.sql
│     └─ sanity_checks.sql
├─ data_preprocessing/
└─ data/
   ├─ public/
   ├─ private/
   └─ dump_folder/
```

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├─ data_preprocessing/
└─ data/
   ├─ public/
   ├─ private/
   └─ dump_folder/
```

Example: psql command for 01_schema.sql

```
psql -v ON_ERROR_STOP=1 -v schema=$SCHEMA -f sql/01_schema.sql "host=$DB_HOST
port=$DB_PORT dbname=$DB_NAME user=$DB_USER password=$DB_PASSWORD"
```

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```

Make commands for easy setup

00_reset.sql	`make reset-truncate` / `make reset-drop`
01_schema.sql	`make schema`
04_load_from_dir.sql	`make load`
example_queries.sql	`make queries`

Section II: Progress Report

Critical thinking and evaluation

Implementation vs. Proposal

What We Proposed

- *Full redesign of CEU's course information system*
- *Collaboration with CEU stakeholders*
- *Integration of real institutional data*

What Changed & Why

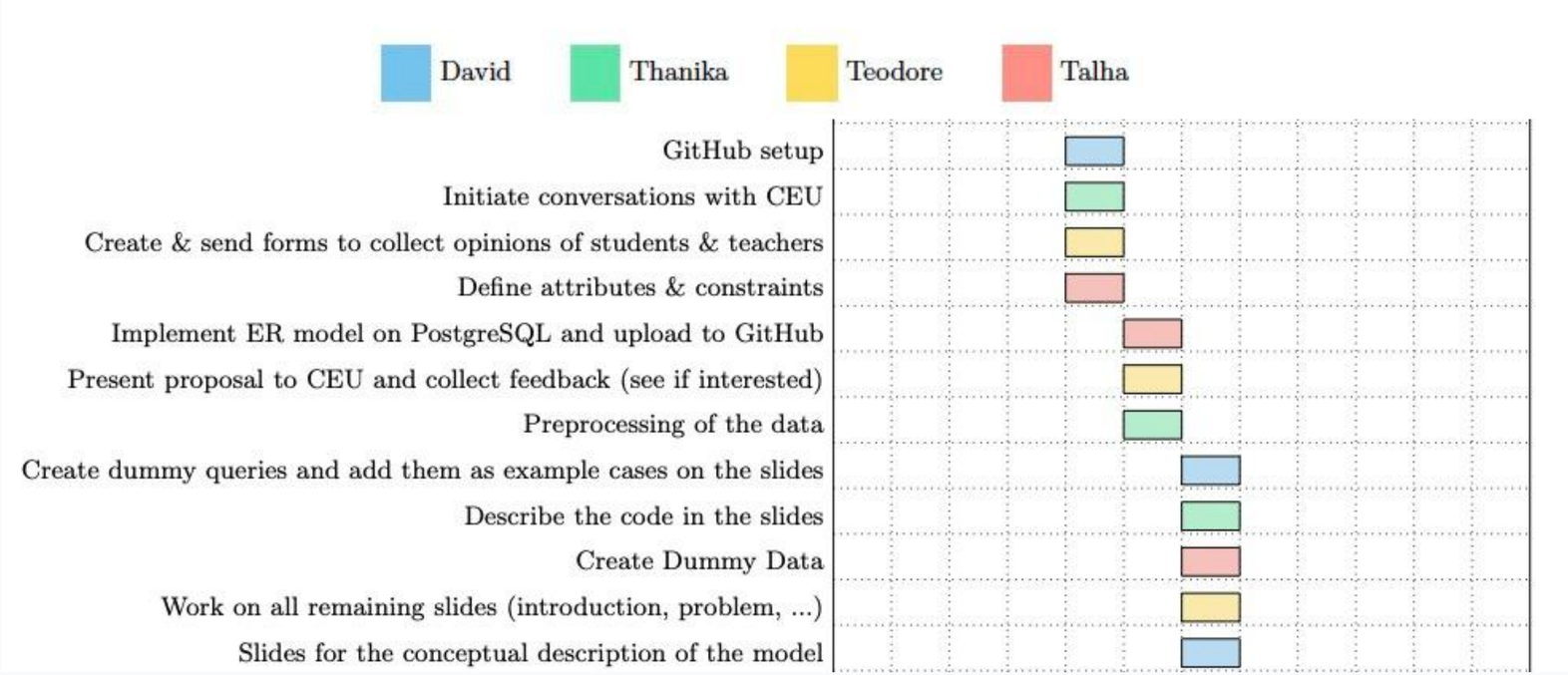
- *Shifted from institutional collaboration → academic prototype*
- *Adjustments based on feedback we have received from faculty*
- *Attributes 'Learning Outcomes' for tables 'Courses' and 'Programs'*
- *Helps program Directors with creating courses*

In Essence...

Why

- *Simply a model database – what we think CEU should implement*
- *fake Scheduling data*
- *fake classrooms*
- *fake lessons*
- *only selected courses/programs with real LO. Fake LOs for others*

Timeline Assessment



On Track

- *Creating the database, populating it with fake data*
- *Preprocessing*
- *Coming up with Queries*
- *No Major Delays (Munther's Comment)*

Delayed / Adjusted

- *Working with CEU for the project (Dropped)*
- *Adding LOs*

Best Practices & Lessons Learned

What We Learned

- *Scope Management: Ambitious goal of engaging with CEU.*
- *Ensure everyone is in the loop, quicker to find flaws in proposed solutions.*
- *Start with dummy data earlier*
- *Validate ER model with test queries immediately*

Reviewer Feedback

- *Added learning outcomes*
- *Clarified use cases*
- *Improved schema detail beyond conceptual level*

Section III: Future Plan

The road ahead — next 4 weeks

New Additions & Updated Goals

Current Feasible Solutions

- *Adjustments based on feedback we have received from faculty*
  - *Attributes 'Learning Outcomes' for tables 'Courses' and 'Programs'*
  - *Helps program Directors with creating courses*
- *add real CEU course information thats available*
- *model database with fake data on what is needed to make a better database*
  - *fake Scheduling data*
  - *fake classrooms*
  - *fake lessons*
  - *only selected courses/programs with real LO. Fake LOs for others*

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  - *fake classrooms*
  - *fake lessons*
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! Lack of Data !

Exploring the CEU Course Data

	Course code	Course code.1	Course code.2	Offered	Taught	Allow repeats	US credits	Instructor	Level	Corresponding programs	...	Course late changes (Spring term)	Course late changes	Course details
0	ECBS6011	Advanced Econometrics 1	Fall term	1.0	Yes	No	2.0	Adam Reiff (reiffa@ceu.edu)	Doctoral or equivalent	PhD in Economics, MA in Economics, Data, and P...	...	NaN	NaN	Maintain
1	ECBS6255	Theoretical Econometrics	Fall term	1.0	Yes	No	2.0	Adam Reiff (reiffa@ceu.edu)	Doctoral or equivalent	MA in Economics, PhD in Economics, MS in Socia...	...	NaN	NaN	Maintain
2	GENS5070	Internship Analysis Workshop	Fall term	1.0	Yes	No	2.0	Adriana Qubaiova (qubaiovaa@ceu.edu)	Master or equivalent	MA in Critical Gender Studies	...	NaN	NaN	NaN
3	GENS5119	Academic Writing Part 2: Thesis Development	Winter term	1.0	Yes	No	1.0	Adriana Qubaiova (qubaiovaa@ceu.edu)	Master or equivalent	MA in Women's and Gender Studies (GEMMA), MA i...	...	NaN	NaN	Approved
4	GENS5564	Gender and Sexuality in West Asia & North Africa	Fall term	1.0	Yes	No	4.0	Adriana Qubaiova (qubaiovaa@ceu.edu)	Master or equivalent	PhD in Comparative Gender Studies, MA in Natio...	...	NaN	NaN	Approved

```
Combined dataframe shape: (1929, 102)
```

```
Total NaN values in combined dataframe: 110067
```

56% of the data is empty

Filtering Useful Columns

From 102 columns to 27.
Some of these will be added to the schema.

Non-Normal Form Data

Column ‘Corresponding_programs’:

- 'PhD in Economics, MA in Economics, Data, and Policy (1 year) - Research Track, MS in Social Data Science (2 years), MA in Economics'
- 'MA in Economics, PhD in Economics, MS in Social Data Science (2 years), MA in Economics, Data, and Policy (1 year) - Research Track'

Column ‘Corresponding_collections’:

- 'ECON 2-year MA 2nd year elective, Economics PhD 1st year mandatory, Mandatory elective courses - Advanced Skills, Specialization module'
- 'MA in Gender Critical Studies 1 of 2 mandatory, MA in Gender Studies year 1 of 1 mandatory, MA in Gender Studies GEMMA 1 of 2 mandatory, MA in Gender Studies GEMMA 2 of 2 elective, MA in Gender Studies MATILDA 1 of 2 mandatory'

Unavailable data

- **Scheduling data**
- **classrooms**
- **lessons**
- **LOs**

Our solution: making fake data and manually populating a selected sample, because this is the purpose of our database.

- **allow students to see complete timetables**
- **Program directors to see the big picture**
- **possibly help administration with rescheduling class**

Plan for the Next 4 Weeks

Week 1

- *Further data cleaning*
- *Turn the data into 2NF*

Week 2

- *Turn data into 3NF*
- *Start Populating the database*
- *!Make changes when needed!*

Week 3

- *Generate fake data:*
  - *fake Scheduling data*
  - *fake classrooms*
  - *fake lessons*
  - *fake learning outcomes*
- *Use real data for a selected sample*
  - *DNDS*
  - *BA DSS*
  - *selected masters program*

Week 4

- *Populate the database*
- *Finalize the database*
- *Make example queries for demonstration*
- *Prepare for the final presentation*

References

[1] E. Lvov and T. Haltrich. CEU Students' experience with course selection process [Survey]. 2025.

[2] Universitat Rovira i Virgili. URV Intranet. 2026. https://intranet.urv.cat/

[3] Holistics Software. dbdiagram.io. 2026. https://dbdiagram.io/

[4] Petra Karl Novak. Suggestion for Learning Outcomes attributes for the tables ‘Course’ and ‘Programs’. 2026.

[5] Central European University. Course Data [Data set]. 2025.

[6] Central European University. TimeEdit Course Data [Data set]. 2026.

[7] T. Giorgobiani. CEU Course Management Experience [Survey]. 2026.